

Viswanathan T J

Software Engineer

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SUMMARY

Software Engineer with **3+ years at Zoho**, building production systems in Python and Java — **REST APIs, microservices**, a shared **API gateway**, and **Kafka**-based cross-service data sync. Ships software end to end: design, development, **Docker**, **CI/CD**, and self-hosted operations.

SKILLS

PROGRAMMING	Python	Java	JavaScript	TypeScript	Shell Scripting						
BACKEND	FastAPI	Flask	Django	REST APIs	Microservices	Kafka	SQLAlchemy				
FRONTEND	React	Next.js	HTML	CSS							
DATABASES	MySQL	PostgreSQL	Redis	Firebase	MongoDB	SQLite					
CLOUD & DEVOPS	Docker	Google Cloud	GitHub Actions	CI/CD	Linux						
TOOLS & SECURITY	Git	Postman	JWT	RBAC	Cryptography						

EXPERIENCE

Software Development Engineer

Jun 2023 – Present

Zoho Corporation · Chennai

- Built a CLI-based **server configuration and deployment automation** tool in **Python and Shell**, adopted by the **entire Zoho Projects engineering team** for setup and releases.
- Cut developer cycle time by **~40%** by streamlining environment setup, code changes, and testing loops.
- Developed backend components for a shared **API gateway** powering **install/uninstall for all Zoho Projects integrations** (10+ external systems) — new integrations are **config-driven**, eliminating **~40% of duplicated integration code**.
- Consolidated **real-time data synchronization** from **5+ services** into one shared design that dispatches **Kafka or scheduler payloads** based on each module's sync requirement, cutting cross-service sync latency **from minutes to seconds** — now the standard pattern for new services.

Software Development Engineer — Intern

Aug 2022 – Jun 2023

Zoho Corporation · Chennai

- Independently built a full-stack **project management tool** — **backend APIs and UI** — adopted by **200+ internal users**.
- Resolved **30+ production issues** in a live product and collaborated on **new-feature analysis** with the core team.
- Drove the team's **Eclipse → IntelliJ migration** by reimplementing slow IDE-plugin workflows as **Python/Shell scripts** with fully automated configuration.

PROJECTS

Academic Suite — Exam Controller

2026

FastAPI · React/TypeScript · PostgreSQL · Redis · Pandas · Docker

- Built a full-stack platform automating **marksheets, hall-ticket, and report generation** for PG/UG/Diploma programs — pixel-accurate PDFs (A4/Legal/A5) from Jinja2 templates, with a unified **grading engine** covering three academic systems via **Strategy + Factory** patterns.
- Served two role-aware React frontends from one **FastAPI** backend: staff portal with granular **RBAC** and time-windowed marks entry; student portal with tab-scoped **JWT** and AES-GCM-encrypted sessions.
- Modeled the data layer on **PostgreSQL** with typed SQLAlchemy 2.0 and a repository pattern; **Dockerized** and self-deployed on a VPS with backups and monitoring.

Agentic AI Platform with MCP Integration

2025

Python · Gemini ADK · MCP · Vertex AI · Cloud Run · A2A Protocol

- Built a Python **multi-agent platform** with **MCP tool integration** and **Gemini ADK**, featuring a news agent that fetches live headlines on demand.
- Implemented **context engineering, LLM-as-judge evaluation**, observability, and security guardrails; deployed on **Vertex AI** and **Cloud Run** with **A2A** workflows.

Marksheet Management System

Jul 2021 – Oct 2021

Flask · Bootstrap · MySQL · Pandas

- Built a **Flask** web app that reads student marks from **Excel sheets** and processes them with **Pandas** into print-ready **marksheets, hall tickets, and department copies**.
- Delivered as a **fully offline, single-machine** system per the institution's requirement — a stable build still running in production today — later evolved into **Academic Suite**, a separate web app accessible from anywhere.

Password Manager — Web & Desktop

Sep 2020 – Dec 2020

Flask · Bootstrap · PyQt5 · SQLite · MongoDB · Firebase

- Built a password manager with **Flask + Bootstrap** (web) and **PyQt5** (desktop) clients, securing credentials with **Fernet symmetric encryption** (Python cryptography) and **hash-based login verification**.
- First end-to-end project — implemented persistence across **SQLite, MongoDB, and Firebase**, learning **CRUD design, encryption, and cross-platform GUI** development.

EDUCATION

MCA — Master of Computer Applications

Kongu Engineering College, Erode · Oct 2021 – Apr 2023

B.Sc Computer Science

Thiagarajar College, Madurai · Apr 2018 – May 2021

CERTIFICATES

RHCSA — Red Hat Certified System Administrator (EX200)

ID: 210-048-773 · 2021

Agent Development (Prototype to Production), LLM Architectures

Google Kaggle · 2025